

Importing the libraries

```
In [2]: import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
```

Importing the dataset

```
In [4]: df = pd.read_csv(r'C:\Users\sande\OneDrive\Desktop\Naresh IT\data science and ai\Ma
df
```

```
Out[4]:
```

	CustomerID	Genre	Age	Annual Income (k\$)	Spending Score (1-100)	k-means
0	1	Male	19	15	39	NaN
1	2	Male	21	15	81	NaN
2	3	Female	20	16	6	NaN
3	4	Female	23	16	77	NaN
4	5	Female	31	17	40	NaN
...	...	...	...	...	...	...
195	196	Female	35	120	79	NaN
196	197	Female	45	126	28	NaN
197	198	Male	32	126	74	NaN
198	199	Male	32	137	18	NaN
199	200	Male	30	137	83	NaN

200 rows × 6 columns

```
In [16]: X = df.iloc[:, [3, 4]].values
```

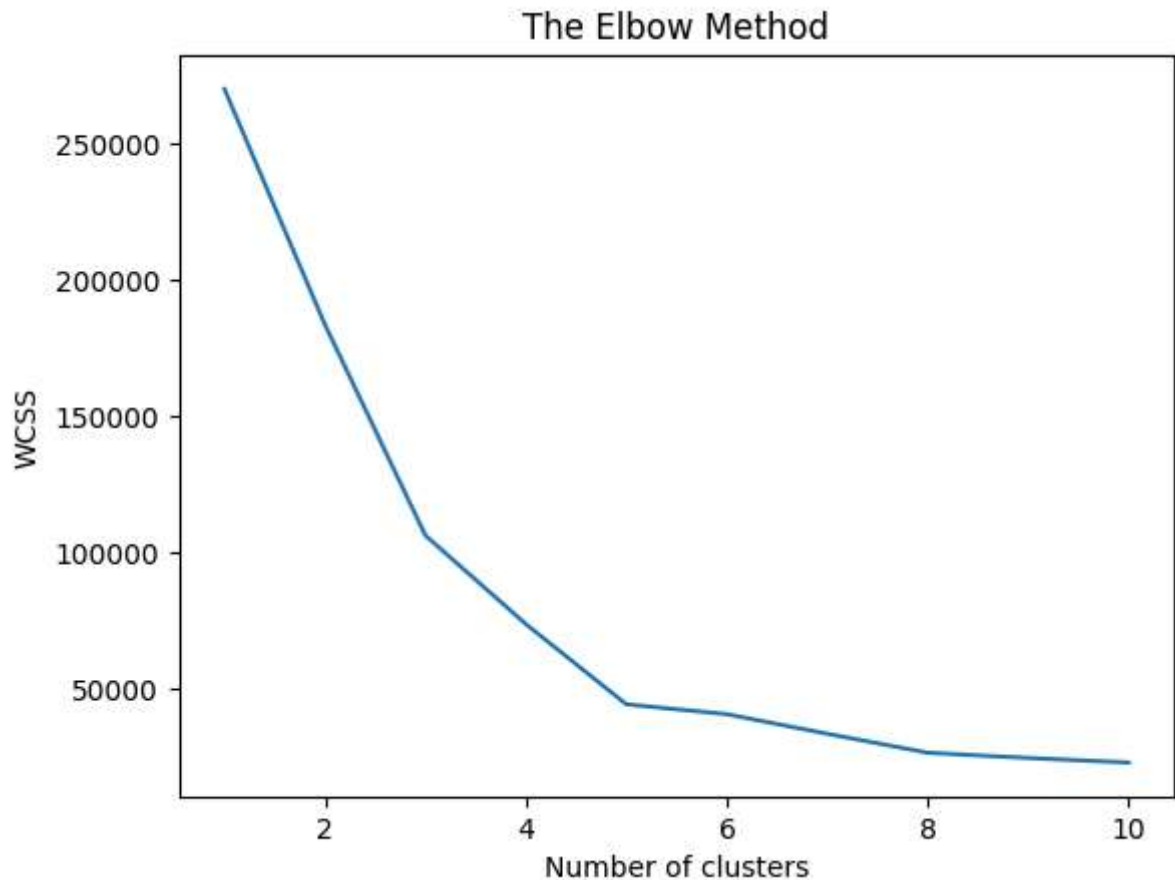
Using the elbow method to find the optimal number of clusters

```
In [17]: from sklearn.cluster import KMeans
import matplotlib.pyplot as plt
```

```
In [18]: wcss = []
for i in range(1, 11):
    kmeans = KMeans(n_clusters=i, init="k-means++", random_state=42)
```

```
kmeans.fit(X)
wcss.append(kmeans.inertia_)
```

```
In [19]: plt.plot(range(1, 11), wcss)
plt.title('The Elbow Method')
plt.xlabel('Number of clusters')
plt.ylabel('WCSS')
plt.show()
```



Training the K-Means model on the dataset

```
In [20]: kmeans = KMeans(n_clusters = 5, init = 'k-means++', random_state = 0)
y_kmeans = kmeans.fit_predict(X)
```

Visualising the clusters

```
In [23]: plt.scatter(X[y_kmeans == 0, 0], X[y_kmeans == 0, 1], s = 100, c = 'red', label = '0')
plt.scatter(X[y_kmeans == 1, 0], X[y_kmeans == 1, 1], s = 100, c = 'blue', label = '1')
plt.scatter(X[y_kmeans == 2, 0], X[y_kmeans == 2, 1], s = 100, c = 'green', label = '2')
plt.scatter(X[y_kmeans == 3, 0], X[y_kmeans == 3, 1], s = 100, c = 'cyan', label = '3')
plt.scatter(X[y_kmeans == 4, 0], X[y_kmeans == 4, 1], s = 100, c = 'magenta', label = '4')
plt.scatter(kmeans.cluster_centers_[:, 0], kmeans.cluster_centers_[:, 1], s = 300, c = 'black')
plt.title('Clusters of customers')
plt.xlabel('Annual Income (k$)')
```

```
plt.ylabel('Spending Score (1-100)')
plt.legend()
plt.show()

df['cluster'] = y_kmeans
```

